

Assignment Code: DA-AG-006

Statistics Advanced - 1| Assignment

Instructions: Carefully read each question. Use Google Docs, Microsoft Word, or a similar tool to create a document where you type out each question along with its answer. Save the document as a PDF, and then upload it to the LMS. Please do not zip or archive the files before uploading them. Each question carries 20 marks.

Total Marks: 200

Question 1: What is a random variable in probability theory?

Answer:

In probability theory, a **random variable** is a mathematical function that assigns a numerical value to each possible outcome in a sample space.

Essentially, it translates "real-world" outcomes (like "Heads" or "Tails") into numbers (like 1 or 0) so they can be analyzed mathematically.

Types of Random Variables

- **Discrete Random Variable:** Takes on a countable number of distinct values (e.g., the number of students in a class).
- **Continuous Random Variable:** Takes on any value within a specific range or interval (e.g., the exact height of a person or the time spent waiting for a bus).

Question 2: What are the types of random variables?

Answer:

There are two main types of random variables:

- **Discrete Random Variables:** These have a countable number of possible outcomes. Examples include the number of heads in coin flips or the number of students in a class.
- **Continuous Random Variables:** These can take any value within a given range or interval. They represent measurements like height, weight, temperature, or time.

Question 3: Explain the difference between discrete and continuous distributions. **Answer:**

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The main difference between discrete and continuous distributions lies in the nature of the random variables they describe.

Comparison Table

Feature	Discrete Distribution	Continuous Distribution
Data Type	Countable values (integers).	Uncountable values within a range (real numbers).
Values	Distinct, separate points (e.g., 0, 1, 2).	Any value within an interval (e.g., 1.527...).
Probability Function	Probability Mass Function (PMF).	Probability Density Function (PDF).
Probability Calculation	$P(X = x)$ is the height at point x .	$P(X = x)$ is always zero ; probability is the area under the curve over an interval.
Graph Style	Isolated bars or dots (Histogram/Bar chart).	Smooth, unbroken curve.

Examples

- **Discrete Distribution:** The number of heads in 10 coin flips, or the number of students in a classroom. You cannot have "2.5 students."
- **Continuous Distribution:** The exact height of people, the time it takes for a battery to die, or the temperature of a room. These can be measured to an infinite number of

decimal places.

Question 4: What is a binomial distribution, and how is it used in probability? **Answer:**

A **binomial distribution** is a discrete probability distribution that models the number of successes in a fixed number of independent trials.

What is it?

It summarizes the probability that a value will take one of two independent values (e.g., Success/Failure, Heads/Tails) under a given set of parameters:

- n : The number of independent trials.
- p : The probability of success in each trial.

The probability of getting exactly k successes in n trials is given by the formula:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

How is it used?

It is used to calculate the likelihood of outcomes in scenarios where there are only two possible results. Common applications include:

- **Quality Control:** Determining the probability of finding a specific number of defective items in a batch.
- **Survey Sampling:** Estimating the likelihood that a certain number of people in a sample will answer "Yes" to a binary question.
- **Finance:** Modeling the probability of a stock price moving up or down over a specific period.
- **Experiments:** Predicting the outcome of repeated coin flips or clinical trials (e.g., whether a patient recovers or not).

Question 5: What is the standard normal distribution, and why is it important? **Answer:**

The Standard Normal Distribution is a specific type of normal distribution where the mean is 0 and the standard deviation is 1. It is often referred to as the Z-distribution.

Key Characteristics

- Notation: $Z \sim N(0, 1)$
- Shape: Symmetric, bell-shaped curve centered at zero.
- Z-score: Any value x from a normal distribution with mean μ and standard deviation σ can be converted to the standard normal distribution using the formula:

$$Z = \frac{x - \mu}{\sigma}$$

Why is it important?

- **Universal Comparison** : It allows researchers to compare data from different scales or populations (e.g., comparing a score on the SAT to a score on the ACT) by converting them into standard "Z-scores."
- **Probability Calculations** : Before modern computers, statisticians used "Z-tables" to find the area under the curve (probability) for any normal distribution. By standardizing data, they only needed one table to calculate probabilities for any set of data.
- **Hypothesis Testing** : It is the foundation for many statistical tests, such as the Z-test, and is critical for determining confidence intervals and p-values in inferential statistics.
- **Central Limit Theorem** : It is the distribution that the means of large samples approach, regardless of the shape of the original population, making it a cornerstone of probability theory.

Question 6: What is the Central Limit Theorem (CLT), and why is it critical in statistics? **Answer:**

The **Central Limit Theorem (CLT)** is a fundamental principle in probability theory and statistics. It explains why the normal distribution is so prevalent in nature and data science.

What is the Central Limit Theorem?

The CLT states that if you take sufficiently large random samples from any population with a finite level of variance, the **distribution of the sample means** will be approximately **normally distributed** (a bell curve), regardless of the shape of the original population distribution.

Key conditions:

- **Sample Size (n):** Generally, a sample size of $n \geq 30$ is considered sufficient for the theorem to hold.
- **Independence:** Observations must be independent of one another.
- **Identically Distributed:** Samples should come from the same underlying distribution.

Why is it critical in statistics?

The CLT is essentially the "bridge" that allows statisticians to make inferences about a population without knowing its true distribution.

1. **Inference and Hypothesis Testing:** It allows for the use of parametric tests (like Z-tests and t-tests) on data that isn't normally distributed, provided the sample size is large enough.
2. **Predictability:** It enables the calculation of **confidence intervals** and **p-values**, as we can predict how sample means will behave.
3. **Standardization:** Because the distribution of means follows a known normal curve, we can use the **Standard Normal Distribution** to compare different datasets.
4. **Practicality:** In the real world, most populations (like income, dice rolls, or website traffic) are skewed or irregular. The CLT allows us to simplify these complex "real-world" shapes into a manageable mathematical model.

Question 7: What is the significance of confidence intervals in statistical

analysis? **Answer:**

Confidence intervals play a crucial role in statistical analysis by providing a range of values that likely contain an unknown population parameter (like a mean or proportion).

Here is why they are significant:

- **Quantifies Uncertainty:** Unlike a "point estimate" (a single number like a mean), a confidence interval acknowledges that every sample has some degree of error. It tells you how much "wiggle room" there is in your data.
- **Indicates Precision:** The width of the interval is a direct measure of precision. A narrower interval suggests a more precise estimate (often due to a larger sample size), while a wider interval indicates more uncertainty.
- **Relates to Statistical Significance:** If a confidence interval for the difference between two groups does not include zero, it typically suggests that the difference is statistically significant.
- **Long-term Reliability:** A 95% confidence level means that if you were to repeat the experiment many times and calculate an interval each time, approximately 95% of those intervals would capture the true population parameter.

Question 8: What is the concept of expected value in a probability

distribution? **Answer:**

The expected value ($E[X]$ or μ) of a probability distribution is the theoretical long-run average value of a random variable over many repetitions of an experiment.

It represents the center or "balance point" of the distribution.

Key Concepts

- **Weighted Average:** Unlike a simple average, the expected value weights each possible outcome by the probability of it occurring.
- **Discrete Formula:** For a discrete random variable, it is the sum of each outcome (x) multiplied by its probability (P(x)):

$$E[X] = \sum x \cdot f(x)$$

Continuous Formula: For a continuous random variable with a probability density function $f(x)$:

$$E[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

Simple Example

If you have a 50% chance of winning \$10 and a 50% chance of winning \$0, the expected value is:

$$(10 \times \text{times } 0.5) + (0 \times \text{times } 0.5) = \$5$$

This doesn't mean you will win \$5 in a single game; rather, it means if you played the game thousands of times, your average winnings would be \$5 per game.

Question 9: Write a Python program to generate 1000 random numbers from a normal distribution with mean = 50 and standard deviation = 5. Compute its mean and standard deviation using NumPy, and draw a histogram to visualize the distribution. *(Include your Python code and output in the code box below.)*

Answer:

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Here is the Python code to generate the distribution, calculate statistics, and visualize the data using NumPy and Matplotlib.

[Code link](#)

Question 10: You are working as a data analyst for a retail company. The company has collected daily sales data for 2 years and wants you to identify the overall sales trend.

daily_sales = [220, 245, 210, 265, 230, 250, 260, 275, 240, 255,

235, 260, 245, 250, 225, 270, 265, 255, 250, 260]

- Explain how you would apply the Central Limit Theorem to estimate the average sales with a 95% confidence interval.
- Write the Python code to compute the mean sales and its confidence interval.

(Include your Python code and output in the code box below.)

Answer:

To address this data analysis task, here is the explanation of the theory and the Python implementation.

Applying the Central Limit Theorem (CLT)

The Central Limit Theorem states that if you have a sufficiently large sample size, the sampling distribution of the mean will be approximately normally distributed, regardless of the population's original distribution.

To estimate the average sales with a 95% confidence interval, follow these steps:

1. **Calculate the Sample Mean ($\bar{x}$)** : This is your point estimate for the population mean.
2. **Calculate the Standard Error (SE)**: Divide the sample standard deviation (s) by the square root of the sample size (n): $SE = \frac{s}{\sqrt{n}}$.
3. **Determine the Z-score**: For a 95% confidence level, the critical value (Z^*) is approximately 1.96.
4. **Compute the Margin of Error** : Multiply the Z-score by the SE.
5. **Find the Interval** : The range is $\bar{x} \pm (Z^* \times SE)$.

[Code link](#)